Experiment 04 — Clustering comparison

arxivkg · preprocessing · phase 5 (topic tree, offline-browse artifact)

What we did

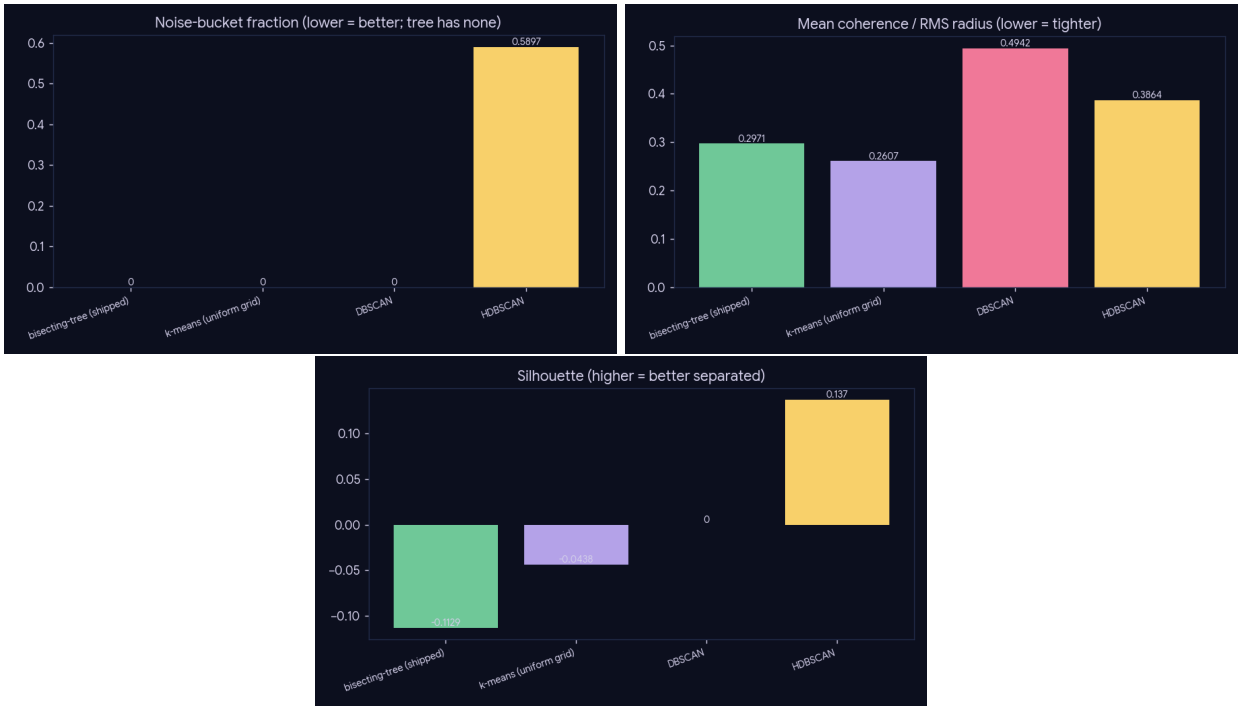
The shipped semantic index is a density-aware **bisecting tree** (`paper_topic`): every paper gets a leaf, coherence early-stop, **no noise bucket**. We compare it on a common sample against a uniform k-means grid and DBSCAN/HDBSCAN, to quantify the trade-off it was built to win — avoiding the noise bucket that swallows niche papers into "Miscellaneous".

Method

Sample 30,000 papers carrying a shipped leaf (768-d nomic → PCA-50); run each clusterer; score noise fraction, cluster count, coherence (RMS radius, lower tighter), silhouette. Sample + labels cached to `cache/summary.json`.

Results (sample n = 30,000, tree leaves in sample = 3,139)

method	noise	clusters	coherence	silhouette
bisecting-tree (shipped)	0.0%	3,139	0.2971	-0.1129
k-means (uniform grid)	0.0%	3,132	0.2607	-0.0438
DBSCAN	0.0%	1	0.4942	0.0
HDBSCAN	59.0%	2	0.3864	0.137



Conclusion

The decisive axis is the **noise bucket**: HDBSCAN dumps **59 %** of the sample into noise (collapsing to 2 clusters); DBSCAN degenerates to a single cluster at this grain. Those discarded low-density points are exactly the niche papers the index must still place. The bisecting tree assigns *every* paper a leaf (0 % noise) at a fine, density-adaptive grain (3,139 leaves); k-means matches the no-noise property only by imposing a uniform grain the tree's coherence early-stop avoids. Silhouette is the wrong lens here (3,000+ fine clusters in PCA-50 are not linearly separated) — the noise-bucket and density-adaptive grain are the structural points, and they are grain-robust. Sampled + PCA-50 proxy, not the full UMAP-50 build.